

PRANAV CHOPADE

MECHANICAL ENGINEERING STUDENT

📞 8999803353
✉ chopadepranav01@gmail.com
🏠 Pune, India
🌐 [LinkedIn](#)

SUMMARY

Mechanical Engineering student with hands-on experience in drivetrain development, CAD modelling, fabrication, and testing through Formula Student projects. Skilled in automotive design, engineering analysis, and cross-functional teamwork, with research contributions in drivetrain systems.

PROJECTS

Custom Limited Slip Differential - Design & Testing

📅 07/2025 - Present

Final Year Project focused on the design and testing of a Custom Limited Slip Differential

- Performed torque bias ratio estimation, gear geometry considerations, and material selection
- Performed FEA and optimized CAD models using SolidWorks

Sheet Metal Upright - Design & Fabrication

📅 01/2025 - 04/2025

Third Year Project for designing and manufacturing a Sheet Metal Upright

- Engineered an upright using 4 mm 7075 aluminium sheet, achieving weight reduction
- Created a puzzle-style layout that simplified laser cutting, welding, and assembly

Oil Skimmer - Design & Fabrication

📅 01/2023 - 05/2023

Diploma Third Year Project focused on developing a simple oil removal mechanism

- Developed and built a small device to separate oil from water
- Integrated a rotating plate mechanism for oil separation

Pipe Master 360 - Freelance Project

📅 03/2025 - 04/2025

Project focused on developing a practical solution for handling heavy industrial pipes

- Designed and manufactured a pipe-handling trolley for construction sites
- Enabled safe and easy movement of heavy fire-fighting pipes

Salon Chair Attachment Tray - Freelance Project

📅 08/2025 - 12/2025

Project focused on improving salon workspace organization and efficiency

- Developed and manufactured an adjustable and detachable tray for salon chairs

EDUCATION

Bachelor of Engineering (Mechanical)

Sinhgad College of Engineering

7.78 CGPA

📅 2023 - 2026

Diploma in Mechanical Engineering

Government Polytechnic, Aurangabad

75.35%

📅 2020 - 2023

PUBLICATIONS

"Design, Analysis and Development of a Custom Helical Limited Slip Differential for Formula Student Applications"

Published in International Journal of Engineering Research & Technology (IJERT), Vol. 15 Issue 05, May 2026

SKILLS

- CAD Modelling (SolidWorks)
- Engineering Documentation
- GD&T Basics
- Design Calculations
- Sheet Metal Fabrication
- Team Management
- Project Planning

EXTRACURRICULAR

Vice-Captain - STES Racing

📅 2024 - 2026

Managed drivetrain design, testing, and coordination activities for the Formula Student Team, while coordinating technical activities and representing the team at Formula Bharat 2025 and Formula Bharat 2026.